

Dhriti Nagar, PhD

Postdoctoral Research Scholar

CONTACT

Department of Pediatrics
Division of Neonatal &
Developmental Medicine
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RESEARCH EXPERIENCE

Postdoctoral Research Scholar

Paşca Lab, Stanford University, Dept.
of Pediatrics · 2022–present

Doctoral Researcher

Ghose Lab, IISER Pune, India ·
2015–2021

Technical Research Assistant

IISER Pune COVID Sequencing
Project, IISER Pune, India · 2020–
2021

Indo-US GETin Fellow (predoctoral)

NHGRI, NIH, Bethesda, MD, USA ·
2018

Summer Undergraduate Research Fellow

ACBR, New Delhi, India · 2013

EDUCATION

PhD, Neuroscience

IISER Pune, India · 2021
Supervisor: Aurnab Ghose, PhD

MS, Biology

IISER Pune, India · 2015

B.Sc (Honours), Biochemistry

University of Delhi, India · 2013

LANGUAGES

English, Hindi

RESEARCH FOCUS

I am a developmental neuroscientist and stem-cell biologist building human, patient-relevant models of the developing brain to uncover disease mechanisms and advance therapeutics. My program spans the genetic and environmental determinants of neuronal development and translational drug discovery.

Modeling environmental and genetic disorders of the developing brain in human organoids. Using human brain organoids and forebrain assembloids, I model how neuronal migration and maturation are disrupted by **environmental** insults (hypoxic-ischemic injury of prematurity) and **genetic** risk (22q11.2 deletion syndrome). I live-imaged interneuron migration under hypoxia and defined mitochondrial fragmentation, impaired oxidative phosphorylation, and dysregulated calcium signaling in 22q11.2DS interneurons, identifying adrenomedullin (ADM) as a common rescue acting through PKA/DRP1 and PLC/calcium pathways.

Engineering next-generation human brain organoid models (NAMs). I develop orthogonally regulated, multi-lineage brain-region organoids and forebrain assembloids (the patent-pending ORChID + INVIGOR platform) that recreate the neuron–glia, neuron–microglia, and neuron–vasculature milieu of the human fetal and neonatal brain for disease modeling and therapeutic testing.

Novel therapeutics: adrenomedullin (ADM) analogs. I am advancing ADM-analog peptide therapeutics from organoid-based discovery toward the clinic (issued and pending patents), and building high-throughput screening pipelines to identify additional neuroprotective compounds.

Previously: cytoskeletal regulation of neural circuit development. My doctoral work established the Formin Fmn2/Fmn2b as a regulator of axon outgrowth, branching, and circuit assembly in zebrafish, and generated the first CRISPR Fmn2b mutants as a community resource.

SKILLS & EXPERTISE

- **New Approach Methodology (NAM) development:** human iPSC culture, brain organoids & forebrain assembloids.
- **Functional genomics:** CRISPR/CRISPRa knockout & activation, prime editing.
- **High-throughput screening (HTS):** assay pipeline development.
- **Drug discovery & development:** peptide therapeutics (ADM analogs), target ID & validation.
- **In silico drug discovery:** AlphaFold structure prediction & molecular docking.
- **Single-cell & multi-omics analysis:** scRNA-seq, transcriptomics, proteomics, metabolic flux.
- **AI/ML-assisted quantification** of imaging & omics data.
- **Advanced microscopy:** live confocal, two-photon Ca^{2+} , light-sheet, SEM.
- **Functional & metabolic phenotyping:** Seahorse, JC-1, intracellular Ca^{2+} , HD-MEA electrophysiology.
- **Human disease modeling:** hypoxic-ischemic injury of prematurity, 22q11.2 deletion syndrome.
- **Translational research & IP:** patents, team science, mentoring.

PATENTS

- **Adrenomedullin Analogs and Methods of Use Thereof.** US20260077017A1 (US 19/283,000); non-provisional filed Jul 28, 2025 (provisional Jul 29, 2024). Co-inventors: A. M. Paşca, Y. Hori, A. Puno. **Published & pending.**
- **Orthogonally Regulated Chemical Induced Differentiation (ORChID) & In Vitro Genetic Switch-Based Organoid Reconstitution (INVIGOR) for Brain Region-Specific Organoids & Assembloids.** PCT filed Jan 28, 2026 (provisional 63/750,437, Jan 28, 2025). Co-inventor: A. M. Paşca. **PCT international pending.**

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• **Antisense Oligonucleotides for Down Syndrome and Tauopathies.** US provisional 63/966,022; filed Jan 22, 2026. Co-inventors: A. M. Paşca, J. B. Choi. **Provisional pending.**

PUBLICATIONS & PREPRINTS

Nagar D, Choi JB, Hu Z, House MS, Park S, Abel B, Nikhil A, Aich M, Htun S, Wang W, Weinacht KG, Gu Z, Paşca AM. **Adrenomedullin restores mitochondrial bioenergetics and rescues interneuron phenotypes in human models of 22q11.2 deletion syndrome.** **bioRxiv**, 2026 (preprint). doi:10.64898/2026.05.24.726075.

Puno A, Michno WP, Li L, Everitt A, McCluskey K, Htun S, **Nagar D**, Choi JB, Dai Y, Park S, Gurwitz E, Willsey AJ, Birey F, Paşca AM. **Adrenomedullin restores the human cortical interneurons migration defects induced by hypoxia.** **eLife**, 2026;14:e108134. doi:10.7554/eLife.108134.

Li L, Choi JB, Shin CH, Htun S, Mestan S, Voss A, Shadrach JL, Puno A, **Nagar D**, Ramirez N, Rojo D, Lee SH, Gibson EM, Kaltschmidt JA, Sloan SA, Chung WS, Paşca AM. **Hypoxia disrupts circadian rhythms in astrocytes and causes synapse engulfment defects.** **bioRxiv**, 2024 (preprint). doi:10.1101/2024.02.22.581651.

Niveditha D, Khan S, Khilari A, Nadkarni S, Bhalerao U, Kadam P, **Nagar D**, et al. **A tale of two waves: diverse genomic and transmission landscapes over 15 months of the COVID-19 pandemic in Pune, India.** **medRxiv**, 2022 (preprint). doi:10.1101/2022.11.05.22281203.

Kundu T, Siva Das S, Sewatkar LK, Kumar DS, **Nagar D**, Ghose A. **Antagonistic activities of Fmn2 and ADF regulate axonal F-actin patch dynamics and the initiation of collateral branching.** **Journal of Neuroscience**, 2022;42(39):7355–7369. doi:10.1523/JNEUROSCI.3107-20.2022.

Ghosh U*, Kundu J*, Ghosh A, Das A, **Nagar D**, Ghose A, Sinha S. **Synthesis of Phosphorodiamidate Morpholino Oligonucleotides using Trityl and Fmoc chemistry in an automated oligo synthesizer.** **The Journal of Organic Chemistry**, 2022;87(15):9466–9478.

Nagar D, James TK, Mishra R, Guha S, Burgess SM, Ghose A. **The Formin Fmn2b is required for the development of an excitatory interneuron module in the zebrafish acoustic startle circuit.** **eNeuro**, 2021;8(4):ENEURO.0329-20.2021.

Kundu T*, Dutta P*, **Nagar D**, Maiti S, Ghose A. **Coupling of dynamic microtubules to F-actin by Fmn2 regulates chemotaxis of neuronal growth cones.** **Journal of Cell Science**, 2021;134(13):jcs252916.

Nagar D, Carrington B, Burgess SM, Ghose A. **Development of motor neurons and motor activity in zebrafish requires F-actin nucleation by Fmn2b.** **bioRxiv**, 2021 (preprint; under revision, PLOS Genetics). doi:10.1101/2021.08.10.455777.

Pathak M, Das R, **Nagar D**, Khan S, et al. **Clinical characteristics of AY.4 infections are similar to B.1.617.2 infections: a preliminary study.** **Indian J. Basic Appl. Med. Res.**, 2021.

* equal contribution. Bold author = D. Nagar.

AWARDS, FELLOWSHIPS & GRANTS

- **Stanford Bio-X Center for Biological Microfluidics (SCBM) C-ShaRP Experiential Learning Program Award** (2026).
- **Outstanding Lightning Talk Award**, Stanford Department of Pediatrics Research Retreat (17th Annual), 2026.
- **Stanford Postdoc JEDI (Justice, Equity, Diversity & Inclusivity) Champion Award** (2024).
- **MCHRI (Maternal and Child Health Research Institute, Stanford University) Postdoctoral Fellowship** (2024).
- **Team Science & Translational Medicine course** for developing neuroprotective therapeutics, MCHRI Stanford, Asilomar, CA (2024).
- **EMBO Travel Grant & Oral Presentation Award**, EMBO Meeting on Modeling Development & Disease with Human Tissue Organoids, Bangalore, India (2023).
- **Pathways to Neuroscience Trainee**, Stanford University (2022).
- **IISER Pune Appreciation Award** for service to the COVID-19 Testing Centre (2022).

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B.Sc (Honours), Biochemistry

University of Delhi, India · 2013

LANGUAGES

English, Hindi

- **WLF Young Investigator** (poster), 4th World Laureate Forum, Shanghai, IBRO (2021).
- **Best Poster Award**, M2T2 Meeting, NBRC Manesar (2019); **DBT-GOI Travel Award**, CSHL Zebrafish Neural Circuits & Behaviour, USA (2019).
- **Indo-US GETin Predoctoral Fellowship** (IUSSTF & DBT-GOI), NIH, USA (2018).
- **Best Poster Award**, InSDB Meeting, CCMB Hyderabad (2015).
- **CSIR JRF (2015–17) & SRF (2017–20)**, India.
- **Silver Jubilee Merit Scholarship**, Dr. Nirmal Tandon Award & Proficiency Award (top of class), B.Sc. Biochemistry, University of Delhi (2010–2013).

INVITED TALKS & PRESENTATIONS

- “Brain Organoid NAMs for CNS Drug Discovery: A Stanford Platform for Disease Modeling and Therapeutic Testing.” **Stanford Drug Discovery Symposium (SDDS)**, Stanford University, CA (2026).
- “Functional and Metabolic Dissection of Inhibitory Neuron Dysfunctions in 22q11.2DS: From Mechanisms to Potential Therapeutics.” **Stanford Department of Pediatrics Research Retreat** (17th Annual), Stanford, CA (April 2026).
- “Mitochondrial Dysfunction in 22q11.2DS Neuropathology.” **9th Biennial Molecular Psychiatry Meeting**, Kona, HI (2024).
- “Mitochondrial Dysfunction in 22q11.2 Deletion Syndrome Neuropathology.” **4th Biennial 22q11 Deletion Syndrome Symposium**, Stanford University, CA (2023).
- **EMBO Meeting on Modeling Development & Disease with Human Tissue Organoids**, Bangalore, India (2023).
- **FENS-Hertie Winter School on Neuroimmune Interactions**, Obergurgl, Austria (2023).
- **No Garland Neuroscience Meeting**, IISER Pune, India (2020).
- **Indo-US GETin Conclave**, NIPGR, New Delhi, India (2020).
- **Monsoon Brain Meeting**, India, virtual (2020).
- **Zebrafish Investigators’ Meeting**, Alibaug, India (2016).
- **Young Asian Developmental Biologists’ Symposium**, International Congress on Zoology, Okinawa, Japan (2016).

SERVICE, OUTREACH, LEADERSHIP & MENTORING

- Co-chair, **Stanford LGBTQ+ Postdocs Group** (SURPAS affinity group), 2022–present.
- Invited participant, **SynBioBeta Genetic Agency Leadership Luncheon**, San Jose, CA (2026).
- Public science talk, “Modeling brains in a dish for personalized medicine,” **Pint of Science**, Palo Alto, CA (May 2026).
- Mentor, **MCHRI DRIVE**, **Stanford SPAN**, and **PIPS** programs for under-represented minorities in science (2022–present).
- Co-founder, **Graduate PodCast Reporter (GPCR)** science-communication series, IISER Pune (2018).
- **Teaching Assistant**, Neurobiology, Evolution & Ecology, Systems Biology, IISER Pune (2015–17); Teacher, **B4 Bioimaging Workshop** (Building Bharat–Boston Biosciences), 2019.
- **Student representative**, Internal Complaints Committee (POSH), IISER Pune (2018–21); Judge & reporter, **NIH PostBac Poster Day / Career Symposium** (2018).
- **Science street plays** & volunteer teaching, Disha NGO, Pune (2013–16); **COVID-19 Testing Centre** volunteer, IISER Pune (2020).

REFERENCES

Anca M. Paşca, MD, Tashia and John Morgridge Endowed Faculty Scholar in Pediatric Translational Medicine; Assistant Professor of Pediatrics, Division of Neonatal & Developmental Medicine, Stanford University School of Medicine (postdoctoral mentor). apasca@stanford.edu · BrainTherapeuticsLab.org

Aurnab Ghose, PhD, Professor, Department of Biology, IISER Pune, India (PhD supervisor); Neuronal Connectivity Lab. aurnab@iiserpune.ac.in · [Neuronal Connectivity Lab](http://NeuronalConnectivityLab)